CEBU INSTITUTE OF TECHNOLOGY - UNIVERSITY
College of Computer Studies, IT411 Capstone & Research 2

Final User & Stakeholder Validation Instrument Package

Deliverable 04: Instruments A (Child), B (Teacher), C (Telemetry), D (Scenarios), E (Parent SUS)

Course & Section:	IT411 (Section G7) — Capstone & Research 2 | Semester 1, AY 2026–2027
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Deliverable Reference:	Pre-Adviser Presentation Requirement D (`Final_Validation_Instrument`) & Week 12 Deliverable 2 (`02_Instruments_and_Data_Collection`, Due October 31, 2026)

Overview of the 5-Instrument Suite

To ensure that **every approved SMART goal is measured using an appropriate empirical tool**—and that system latency, speech recognition accuracy, child task completion, and offline persistence are measured through direct observation, blind acoustic rating, and system telemetry rather than opinion surveys—this package contains five linked instruments:

Instrument Code	Instrument Title	Target Evaluator / Data Source	Measured SMART Goals & Sub-Objectives
Instrument A	**Teacher Phoneme, Decodable Word & Pedagogical Audit Form**	DepEd Grade 1 Reading Teachers (N = 4–5)	`SMART-01`, `SMART-02`, `HI-1`, O6 (Gate 3 acoustic purity & curriculum alignment)
Instrument B	**Child Session Silent-Observer Sheet & Blind Teacher-Rater `Say It` Log**	Silent Observer + 2 Blind Teacher Raters during Child Sessions (N = 16–20 children)	`SMART-02`, `SMART-03`, `SMART-04`, `SMART-05`, `HI-2`, `SI-1`–`SI-4`, O3
Instrument C	**Automated App Telemetry & Room SQLite Extraction Protocol**	Local Device SQLite Database (`playit_database`) & PDF Report	`SMART-03`, `SMART-04`, `SMART-05`, `SMART-07` (Objective attempt counts, accuracy, stars, retention)
Instrument D	**Technical, Performance & Accessibility Verification Checklist (`D-01`–`D-12`)**	Technical / QA Evaluators (N = 3) on Samsung Galaxy A21s (`SM-A217F`)	`SMART-03`, `SMART-04`, `SMART-06`, O3, O4, O5 (Latency, 60 FPS, 20× force-close, Airplane Mode)
Instrument E	**Post-Session Child Picture Choice & Adult Stakeholder Survey (SUS + TAM)**	Part E1: Grade 1 Learners (N = 16–20) Part E2: Teachers & Parents (N = 12–15)	`SMART-05`, `SMART-07`, O6 (Child affect/replay, Adult task completion, Brooke SUS, TAM)

Instrument A: Teacher Phoneme, Decodable Word & Pedagogical Audit Form

Evaluator Code: `T-____` | **Years Teaching Grade 1 Reading:** `____` | **Date:** `2026-10-____`
App Build Verified: `playIT v1.2.0-rc2`

Part A1 — Gate 3 Blind Phoneme Audio Audit (26 Isolated Phoneme Clips: `SMART-01`, `SMART-02`, `HI-1`)

Instructions for Facilitator: Play each clip from `audio/phonemes/{letter}.mp3` in randomized order (`Clip #01` to `Clip #26`) without showing the screen.
Instructions for Teacher: Write the letter sound you hear, check **Yes** if any unwanted *"uh"* (/■/ schwa) vowel is attached to a consonant, and rate acoustic quality (1 = Reject / Mispronounced, 2 = Acceptable for Grade 1, 3 = Exemplary Model).

Randomized Clip #	Actual File Key (Facilitator Key)	Target IPA & Articulatory Class	Teacher Blind Transcription (What letter sound did you hear?)	Schwa /■/ Intrusion Heard? (`Yes` / `No`)	Clarity & Purity Rating (`1`, `2`, or `3`)	Gate 3 Pass? (`Transcription Match` + `No Schwa` + `Rating >= 2`)
A1-01	`m.mp3` (Grp 1)	/m/ (Bilabial nasal, continuous)	`[ ]`	`[ ] Yes [ ] No`	`[ ] 1 [ ] 2 [ ] 3`	`[ ] PASS [ ] FAIL`
A1-02	`s.mp3` (Grp 1)	/s/ (Alveolar fricative, continuous)	`[ ]`	`[ ] Yes [ ] No`	`[ ] 1 [ ] 2 [ ] 3`	`[ ] PASS [ ] FAIL`

Randomized Clip #	Actual File Key (Facilitator Key)	Target IPA & Articulatory Class	Teacher Blind Transcription (What letter sound did you hear?)	Schwa /■/ Intrusion Heard? (Yes / No)	Clarity & Purity Rating (1, 2, or 3)	Gate 3 Pass? (Transcription Match + No Schwa + Rating >= 2)
A1-03	a.mp3 (Grp 1)	/æ/ (Short front vowel — *"a, a, apple"*)	[ ]	[ ] Yes [ ] No	[ ] 1 [ ] 2 [ ] 3	[ ] PASS [ ] FAIL
A1-04	i.mp3 (Grp 1)	/■/ (Short high front vowel — *insect*)	[ ]	[ ] Yes [ ] No	[ ] 1 [ ] 2 [ ] 3	[ ] PASS [ ] FAIL
A1-05	o.mp3 (Grp 1)	/■/ (Short open back vowel — *orange*)	[ ]	[ ] Yes [ ] No	[ ] 1 [ ] 2 [ ] 3	[ ] PASS [ ] FAIL
A1-06	b.mp3 (Grp 1)	/b/ (Voiced bilabial stop, clipped)	[ ]	[ ] Yes [ ] No	[ ] 1 [ ] 2 [ ] 3	[ ] PASS [ ] FAIL
A1-07	e.mp3 (Grp 2)	/■/ (Short mid front vowel — *elephant*)	[ ]	[ ] Yes [ ] No	[ ] 1 [ ] 2 [ ] 3	[ ] PASS [ ] FAIL
A1-08	u.mp3 (Grp 2)	/■/ (Short central vowel — *umbrella*)	[ ]	[ ] Yes [ ] No	[ ] 1 [ ] 2 [ ] 3	[ ] PASS [ ] FAIL
A1-09	t.mp3 (Grp 2)	/t/ (Voiceless alveolar stop, clipped)	[ ]	[ ] Yes [ ] No	[ ] 1 [ ] 2 [ ] 3	[ ] PASS [ ] FAIL
A1-10	k.mp3 (Grp 2)	/k/ (Voiceless velar stop, clipped)	[ ]	[ ] Yes [ ] No	[ ] 1 [ ] 2 [ ] 3	[ ] PASS [ ] FAIL
A1-11	l.mp3 (Grp 2)	/l/ (Alveolar lateral approximant)	[ ]	[ ] Yes [ ] No	[ ] 1 [ ] 2 [ ] 3	[ ] PASS [ ] FAIL
A1-12	y.mp3 (Grp 2)	/j/ (Palatal approximant — *yoyo*)	[ ]	[ ] Yes [ ] No	[ ] 1 [ ] 2 [ ] 3	[ ] PASS [ ] FAIL
A1-13	n.mp3 (Grp 3)	/n/ (Alveolar nasal, continuous)	[ ]	[ ] Yes [ ] No	[ ] 1 [ ] 2 [ ] 3	[ ] PASS [ ] FAIL
A1-14	g.mp3 (Grp 3)	/■/ (Voiced velar stop — *gift*)	[ ]	[ ] Yes [ ] No	[ ] 1 [ ] 2 [ ] 3	[ ] PASS [ ] FAIL
A1-15	p.mp3 (Grp 3)	/p/ (Voiceless bilabial stop, clipped)	[ ]	[ ] Yes [ ] No	[ ] 1 [ ] 2 [ ] 3	[ ] PASS [ ] FAIL
A1-16	r.mp3 (Grp 3)	/■/ (Postalveolar approximant — *rabbit*)	[ ]	[ ] Yes [ ] No	[ ] 1 [ ] 2 [ ] 3	[ ] PASS [ ] FAIL
A1-17	d.mp3 (Grp 3)	/d/ (Voiced alveolar stop, clipped)	[ ]	[ ] Yes [ ] No	[ ] 1 [ ] 2 [ ] 3	[ ] PASS [ ] FAIL
A1-18	h.mp3 (Grp 3)	/h/ (Voiceless glottal fricative — *hat*)	[ ]	[ ] Yes [ ] No	[ ] 1 [ ] 2 [ ] 3	[ ] PASS [ ] FAIL
A1-19	w.mp3 (Grp 4)	/w/ (Labio-velar approximant — *watch*)	[ ]	[ ] Yes [ ] No	[ ] 1 [ ] 2 [ ] 3	[ ] PASS [ ] FAIL
A1-20	c.mp3 (Grp 4)	/k/ (Hard C velar stop — *cat*)	[ ]	[ ] Yes [ ] No	[ ] 1 [ ] 2 [ ] 3	[ ] PASS [ ] FAIL
A1-21	f.mp3 (Grp 4)	/f/ (Voiceless labiodental fricative)	[ ]	[ ] Yes [ ] No	[ ] 1 [ ] 2 [ ] 3	[ ] PASS [ ] FAIL
A1-22	j.mp3 (Grp 5)	/d■/ (Voiced palato-alveolar affricate)	[ ]	[ ] Yes [ ] No	[ ] 1 [ ] 2 [ ] 3	[ ] PASS [ ] FAIL
A1-23	q.mp3 (Grp 5)	/kw/ (Labialized velar cluster — *queen*)	[ ]	[ ] Yes [ ] No	[ ] 1 [ ] 2 [ ] 3	[ ] PASS [ ] FAIL
A1-24	v.mp3 (Grp 5)	/v/ (Voiced labiodental fricative)	[ ]	[ ] Yes [ ] No	[ ] 1 [ ] 2 [ ] 3	[ ] PASS [ ] FAIL
A1-25	x.mp3 (Grp 5)	/ks/ (Final cluster as in *box*)	[ ]	[ ] Yes [ ] No	[ ] 1 [ ] 2 [ ] 3	[ ] PASS [ ] FAIL
A1-26	z.mp3 (Grp 5)	/z/ (Voiced alveolar fricative — *zip*)	[ ]	[ ] Yes [ ] No	[ ] 1 [ ] 2 [ ] 3	[ ] PASS [ ] FAIL

Part A2 — Decodable Blend It Word Bank Audit (33 Words Across Groups 1–6: SMART-01, SMART-04)

Verify that every Blend It word uses **only** letters taught up to that group (100% cumulative decodability) and that the spoken word clip and Argon 4 picture are age-appropriate for Grade 1:

Group	Cumulative Letters Taught	Audited Decodable Words (audio/words/word_*.mp3 & images/pictures/blendword_*.png)	Strictly Decodable? (Yes/No)	Audio & Picture Clear for Grade 1? (Yes/No)	Teacher Remarks
Group 1	M, S, A, I, O, B	AM, SAM, AIM, SIS, SOB, BIM *(6 words)*	[ ] Yes [ ] No	[ ] Yes [ ] No	
Group 2	+ E, U, T, K, L, Y	MAT, SAT, SIT, BAT, BET, BIT, TUB, KIT, LIT, YAM *(10 words)*	[ ] Yes [ ] No	[ ] Yes [ ] No	
Group 3	+ N, G, P, R, D, H	MAN, SUN, PIN, PAN, DOG, PIG, HAT, RED, HUG *(9 words)*	[ ] Yes [ ] No	[ ] Yes [ ] No	
Group 4	+ W, C, F	CAT, CUP, FAN, FIN, WET, WEB *(6 words)*	[ ] Yes [ ] No	[ ] Yes [ ] No	
Group 5–6	+ J, Q, V, X, Z	JAM, JET, VAN, VET, BOX, SIX, ZIP *(7 words; 33 active pool)*	[ ] Yes [ ] No	[ ] Yes [ ] No	

Part A3 — Teacher Pedagogical & Curriculum Alignment Scale (SMART-01, 06)

Rate each statement from **1 (Strongly Disagree)** to **5 (Strongly Agree)**:

Item ID	Pedagogical Alignment Statement	Rating (1–5)
PA-01	The 6-group Marungko-inspired letter progression (M, S, A, I, O, B first) aligns with how early phonics and decoding should be sequenced for Grade 1 learners.	[1] [2] [3] [4] [5]
PA-02	The Hear It 3-step sequence (pure phoneme sound → example word → guided prompt) models accurate phonemic awareness without confusing letter names with letter sounds.	[1] [2] [3] [4] [5]
PA-03	The Say It 3-attempt scaffolding (adding the slow phoneme model and synchronized mouth-shape cue on Attempt 2 while never deducting hearts) is developmentally appropriate for hesitant 6-to-7-year-old speakers.	[1] [2] [3] [4] [5]
PA-04	The Find It picture discrimination grid and Blend It CVC tile-building activities effectively reinforce grapheme-phoneme correspondence and early word synthesis.	[1] [2] [3] [4] [5]
PA-05	The Parent/Teacher Dashboard (Green / Yellow / At-Risk flags) and exported A4 PDF report provide actionable diagnostic data for classroom reading remediation.	[1] [2] [3] [4] [5]

Instrument B: Child Session Silent-Observer Sheet & Blind Teacher-Rater Say It Log

Learner Code: C-____ | **Age:** [ ] 6 [ ] 7 | **Gender:** ____ | **Session Date:** 2026-10-____
Silent Observer Code: F-____ | **Blind Teacher Raters:** T-____ & T-____

Part B1 — Screen-by-Screen Silent-Observer Navigation & Task Log (SMART-02, SMART-04, SMART-05, HI-2, 03)

Observer Rule: Do **not** coach or point unless the child explicitly asks for help or is stuck for ≥ 15 s after Lily's 10s idle prompt. Tally every adult prompt.

Task / Stage	Target Action Observed	Completed? (Pass/Fail)	Adult Prompts Needed (0, 1, 2+)	Stage Time (sec)	Behavioral & Ergonomic Notes (e.g., missed tap, hesitation, smile/delight)
1. Splash & Avatar Onboarding	Taps Start → + New Player → selects animal friend → taps Let's Play	[ ] Pass [ ] Fail	[ ]	____ s	
2. Bohol Adventure Map	Locates pulsing active node (M / S / A) and taps to launch lesson	[ ] Pass [ ] Fail	[ ]	____ s	
3. Hear It (Letter 1: ____)	Listens to 3-step sequence (Sound → Word → Prompt), taps Play & Next	[ ] Pass [ ] Fail	[ ]	____ s	
4. Say It (Letter 1: ____)	Taps 96dp mic button, speaks into mic, completes echo & recall steps	[ ] Pass [ ] Fail	[ ]	____ s	Mic hesitation? [ ] None [ ] Mild [ ] Refused
5. Find It (Letter 1: ____)	Completes 3 rounds of picture/letter selection and taps Continue	[ ] Pass [ ] Fail	[ ]	____ s	Hearts remaining: [ ] / 5
6. Letter Complete	Views earned stars (1star–3star) and taps Continue to return to Map	[ ] Pass [ ] Fail	[ ]	____ s	Stars earned: [ ] 1 [ ] 2 [ ] 3

Task / Stage	Target Action Observed	Completed? (Pass/Fail)	Adult Prompts Needed (0, 1, 2+)	Stage Time (sec)	Behavioral & Ergonomic Notes (e.g., missed tap, hesitation, smile/delight)
7. Lesson 2 (____)	Completes Hear It → Say It → Find It for second letter	[ ] Pass [ ] Fail	[ ]	____ s	Stars earned: [ ] 1 [ ] 2 [ ] 3
8. Blend It Milestone	Taps phoneme tiles in order and blends 3 CVC words (AM, SAM, MAT)	[ ] Pass [ ] Fail	[ ]	____ s	Words solved: [ ] / 3

Part B2 — Trial-by-Trial Say It Blind Teacher-Rater vs. Vosk ASR Log (SMART-03, SI-1–SI-3)

For each spoken Say It attempt by learner C-____, the two teachers independently mark whether the child's pronunciation was acceptable for Grade 1 (1 = Correct Target, 0 = Incorrect / Letter Name Foil / Added Schwa / Silence), while the observer records the app's decision (A in {0, 1}) and latency:

Trial #	Target Letter / Word	Attempt # (1, 2, 3)	Child Utterance Type Observed (Target / Letter-Name Foil / Schwa Foil / Other / Silence)	Teacher Rater 1 (T in {0,1})	Teacher Rater 2 (T in {0,1})	Teacher Consensus (T in {0,1})	Vosk ASR Decision (A in {0,1})	Confusion Matrix Class (TP, TN, FP, FN)	Mouth Cue Used on Att. 2/3? (Yes/No/N/A)
01	____	[ ]1 [ ]2 [ ]3	[ ]	[ ]1 [ ]0	[ ]1 [ ]0	[ ]1 [ ]0	[ ]1 [ ]0	[ ]TP [ ]TN [ ]FP [ ]FN	[ ]Y [ ]N [ ]N/A
02	____	[ ]1 [ ]2 [ ]3	[ ]	[ ]1 [ ]0	[ ]1 [ ]0	[ ]1 [ ]0	[ ]1 [ ]0	[ ]TP [ ]TN [ ]FP [ ]FN	[ ]Y [ ]N [ ]N/A
03	____	[ ]1 [ ]2 [ ]3	[ ]	[ ]1 [ ]0	[ ]1 [ ]0	[ ]1 [ ]0	[ ]1 [ ]0	[ ]TP [ ]TN [ ]FP [ ]FN	[ ]Y [ ]N [ ]N/A
04	____	[ ]1 [ ]2 [ ]3	[ ]	[ ]1 [ ]0	[ ]1 [ ]0	[ ]1 [ ]0	[ ]1 [ ]0	[ ]TP [ ]TN [ ]FP [ ]FN	[ ]Y [ ]N [ ]N/A
05	____	[ ]1 [ ]2 [ ]3	[ ]	[ ]1 [ ]0	[ ]1 [ ]0	[ ]1 [ ]0	[ ]1 [ ]0	[ ]TP [ ]TN [ ]FP [ ]FN	[ ]Y [ ]N [ ]N/A
06	____	[ ]1 [ ]2 [ ]3	[ ]	[ ]1 [ ]0	[ ]1 [ ]0	[ ]1 [ ]0	[ ]1 [ ]0	[ ]TP [ ]TN [ ]FP [ ]FN	[ ]Y [ ]N [ ]N/A

Scoring Formulas for Part B2 (SMART-03 / SI-3): - **Teacher-ASR Agreement:** P_(agree) = (TP + TN) / (TP + TN + FP + FN) × 100% *(Target: ≥ 80.0%)* - **False-Reject Rate (FRR):** P_(FR) = FN / (TP + FN) × 100% *(Target: ≤ 15.0%)*

Instrument C: Automated App Telemetry & Room Database Extraction Protocol

To substantiate observational records with objective system logs (SMART-03, SMART-04, SMART-05, SMART-07), the following local SQLite tables in playit_database and the exported PDF report are archived after each validation session:

Room Entity / Table	Extracted Columns	Validation Metric Derived
profiles	id, name, avatarResId, currentStreak, lastActiveDate, createdAt	Onboarding completion, streak increment accuracy (FR-09, FR-10).
say_it_attempts	id, profileId, phonemeId, recognizedText, isCorrect, attemptNumber, timestamp	Empirical Say It attempt count (1–3), pass rate per phoneme, Attempt 2/3 recovery rate (SI-1, SI-2).
find_it_attempts	id, profileId, phonemeId, selectedPhonemeId, isCorrect, roundNumber, heartsRemaining, timestamp	Empirical Find It first-tap discrimination accuracy, hearts lost per lesson (FR-05–FR-08).
blend_it_attempts	attemptId, profileId, wordId, groupId, isCorrect, attemptsTaken, heartsRemaining, timestamp	Empirical Blend It CVC word decoding accuracy and attempts per word (FR-NEW-01, FR-13).
lesson_progress	id, profileId, phonemeId, isUnlocked, isCompleted, starsEarned, bestAccuracy, lastPracticedAt	Star distribution (1star, 2star, 3star), 7-day retention window, Green/Yellow/At-Risk status (FR-09, FR-12).

ADB Extraction Command (Executed by QA Evaluator at End of Testing Day):

```
adb exec-out run-as com.playit.app sqlite3 -header -csv databases/playit_database "SELECT * FROM say_it_attempts;" >
validation_say_it_attempts.csv
adb exec-out run-as com.playit.app sqlite3 -header -csv databases/playit_database "SELECT * FROM find_it_attempts;" >
validation_find_it_attempts.csv
adb exec-out run-as com.playit.app sqlite3 -header -csv databases/playit_database "SELECT * FROM blend_it_attempts;"
> validation_blend_it_attempts.csv
adb exec-out run-as com.playit.app sqlite3 -header -csv databases/playit_database "SELECT * FROM lesson_progress;" >
```

validation_lesson_progress.csv

Instrument D: Technical, Performance & Accessibility Verification Checklist (D-01 to D-12)

Evaluator Code: QA-____ | **Test Device:** Samsung Galaxy A21s (SM-A217F, Android 12, Exynos 850, 3 GB RAM)| **Date:** 2026-10-____

Check ID	Linked Requirement	Verification Procedure & Workload	Expected Acceptance Criterion	Measured Value / Actual Result	Status (PASS/FAIL)
D-01	NFR-Perf-01 (Say It Latency)	Execute N = 30 spoken Say It utterances on SM-A217F; measure t_(feedback) − t_(EOS) (ms).	**P90 ≤ 500 ms**	P50: ____ ms | P90: ____ ms | Max: ____ ms	[ ] PASS [ ] FAIL
D-02	NFR-Perf-03 (Find It Latency)	Execute N = 30 tile taps in Find It; measure tap-to-audio/visual feedback delta (ms).	**P90 ≤ 300 ms**	P50: ____ ms | P90: ____ ms | Max: ____ ms	[ ] PASS [ ] FAIL
D-03	NFR-Perf-02 (Audio Clip Length)	Inspect duration of all 26 audio/phonemes/*.mp3 and 33 audio/words/*.mp3 clips.	**100% of clips ≤ 5.0 s (5000 ms)**	Max phoneme: ____ s | Max word: ____ s	[ ] PASS [ ] FAIL
D-04	NFR-Perf (Cold Start, 60 FPS & RAM)	Run adb shell am start -W (10 ×), dumpsys gfxinfo during map scroll & lesson play, and dumpsys meminfo.	**Cold start ≤ 2500 ms**; **Jank ≤ 10%**; **PSS RAM ≤ 200 MB**	Start: ____ ms | Jank: ____% | RAM: ____ MB	[ ] PASS [ ] FAIL
D-05	NFR-Off-02 (20× Force-Close Persistence)	In Airplane Mode, complete/update progress, execute adb shell am force-stop com.playit.app, relaunch, and verify SQLite state **20 consecutive times**.	**20 / 20 cycles (100%)** retain exact stars, hearts, and streaks with 0 corruption.	____ / 20 cycles intact	[ ] PASS [ ] FAIL
D-06	NFR-Off-01 (Airplane Mode Operability)	With Airplane Mode ON and Wi-Fi/Data OFF, execute a full cycle of Hear It, Say It (Vosk), Find It, Blend It, and PDF Export.	**4 / 4 learning modules + PDF export (100%)** operate with zero network errors.	____ / 4 modules + PDF verified	[ ] PASS [ ] FAIL
D-07	NFR-Acc-01 (≥ 64 dp Touch Targets)	Verify bounds of all child interactive controls across Compact (320x640), A21s (360x800), Phone (393x852), and Tablet (800x1280) via LayoutMatrixTest.	**100%** of child buttons (Play, Mic, Next, Back, MapNode) ≥ 64 dp × 64 dp.	Min measured target: ____ dp x ____ dp	[ ] PASS [ ] FAIL
D-08	NFR-Acc-02 (Contrast, Text Floor & Zero Red)	Audit WCAG contrast ratios, font sizes (>=16sp map, >=24sp lesson), element group count (<=5), and zero #D63031 harsh red / zero emojis (ZeroEmojiPolicyTest).	**Contrast ≥ 4.5:1**; **≤ 5 groups/screen**; **0 harsh red**; **0 emojis**.	Min contrast: ____:1 | Emojis: 0	[ ] PASS [ ] FAIL
D-09	SI-2 (Articulatory Mouth Cues)	Trigger Attempt 2 across all 7 ArticulationGroup categories and verify PictureAssetsTest.everyMouthCueIsReleased.	**7 / 7 (100%)** mouth-shape illustrations render left of the mic without clipping.	____ / 7 mouth cues verified	[ ] PASS [ ] FAIL
D-10	SI-1 (≤ 100 ms Mic State Cue)	Tap 96dp mic button in Say It and measure transition from Idle to Listening visual pulse + chime.	**Mic state transition ≤ 100 ms**.	Measured transition: ____ ms	[ ] PASS [ ] FAIL
D-11	NFR-Sec-01..03 (Privacy & Manifest)	Inspect merged AndroidManifest.xml (ManifestPrivacyTest) and network traffic during full session.	**0** INTERNET **permissions**; only RECORD_AUDIO declared; **0 outbound packets**.	Permissions found: RECORD_AUDIO only	[ ] PASS [ ] FAIL
D-12	FR-12, FR-13 (PDF & Dashboard Parity + Stress)	Export A4 PDF from ParentDashboardScreen and run 5,000-event adb shell monkey stress test.	**100% metric parity** between Dashboard & PDF; **0 crashes / 0 ANRs**.	PDF size: ____ KB | Crashes/ANRs: 0	[ ] PASS [ ] FAIL

Instrument E: Post-Session Child Picture Choice & Adult Stakeholder Survey (SUS + TAM)

Part E1 — Child Post-Session Exit Sheet (Non-Reading Protocol: SMART-05, 03)

Administered verbally by the facilitator in ≤ 90 seconds at the end of the child's play session (C-____):

1. **Item E1-01 (Visual Smileyometer):** *"Point to the face that shows how you felt playing with Lily today:"*
   - [ ] 3 — Big Smile (Loved it / Happy)
   - [ ] 2 — Neutral Face (Okay)

- [ ] 1 — Frown (Did not like it / Frustrated)

2. **Item** E1-02 **(Picture-Card Hardest Stage Choice):** Show the 4 stage picture cards (Hear It Speaker, Say It Microphone, Find It Picture Grid, Blend It Word Tiles) and ask: *"Which game was the trickiest or hardest for you today?"*
   - [ ] Hear It (Listening)|[ ] Say It (Speaking into Mic)|[ ] Find It (Finding Pictures)|[ ] Blend It (Building Words)|[ ] None — All Easy
3. **Item** E1-03 **(Behavioral Voluntary Replay Prompt):** *"You finished our game! Would you like to play one more letter right now, or stop here?"*
   - [ ] Chose to play another letter (Voluntary Replay = 1)
   - [ ] Chose to stop now (Voluntary Replay = 0)

Part E2 — Adult Stakeholder Questionnaire (Teachers T-____ & Parents P-____: SMART-07, 06)

Respondent Code: [ ] T-____ [ ] P-____ | **Role:** [ ] Grade 1 Teacher [ ] Parent / Guardian | **Date:** 2026-10-____

Section 1: Adult Hands-On Task Verification (Observed Before Survey)

- Task A1 **(Parent Gate):** Unlocked Parent Zone via ArithmeticGuardDialog on first attempt? [ ] Yes [ ] No
- Task A2 **(Profile Management):** Created/renamed learner profile and switched active learner? [ ] Yes [ ] No
- Task A3 **(Diagnostic Interpretation):** Correctly identified which letters are Mastered (Green), Developing (Yellow), and At-Risk (Amber)? [ ] Yes [ ] No
- Task A4 **(Offline PDF Report):** Exported and opened the A4 PDF Progress Report offline? [ ] Yes [ ] No

Section 2: Standardized System Usability Scale (Brooke SUS — 10 Items, 5-Point Likert 1 = Strongly Disagree to 5 = Strongly Agree)

Item ID	SUS Statement	Score (1–5)
SUS-01	I think that I would like to use the **playIT** application frequently for my child's / students' phonics practice.	[1] [2] [3] [4] [5]
SUS-02	I found the **playIT** application unnecessarily complex. *(Reverse-scored)*	[1] [2] [3] [4] [5]
SUS-03	I thought the **playIT** application was easy to use for both children and adults.	[1] [2] [3] [4] [5]
SUS-04	I think that I would need the support of a technical person to be able to use this application. *(Reverse-scored)*	[1] [2] [3] [4] [5]
SUS-05	I found the various functions in **playIT** (Hear It, Say It, Find It, Blend It, and Parent Zone) were well integrated.	[1] [2] [3] [4] [5]
SUS-06	I thought there was too much inconsistency in this application. *(Reverse-scored)*	[1] [2] [3] [4] [5]
SUS-07	I would imagine that most Grade 1 learners, parents, and teachers would learn to use **playIT** very quickly.	[1] [2] [3] [4] [5]
SUS-08	I found the application very cumbersome or awkward to use. *(Reverse-scored)*	[1] [2] [3] [4] [5]
SUS-09	I felt very confident navigating the lessons and interpreting the Parent Zone progress report.	[1] [2] [3] [4] [5]
SUS-10	I needed to learn a lot of things before I could get going with this application. *(Reverse-scored)*	[1] [2] [3] [4] [5]

Section 3: Technology Acceptance Model (TAM) & Diagnostic Value Scale (1 = Strongly Disagree to 5 = Strongly Agree)

Item ID	Construct	Statement	Score (1–5)
TAM-PU1	Perceived Usefulness	**playIT** helps Grade 1 learners recognize and pronounce individual letter sounds (Hear It & Say It) more effectively.	[1] [2] [3] [4] [5]
TAM-PU2	Perceived Usefulness	The Find It and Blend It stages help learners connect letter sounds to pictures and blend sounds into simple CVC words.	[1] [2] [3] [4] [5]
TAM-PU3	Perceived Usefulness	The Parent Zone (Green / Yellow / At-Risk flags and 7-day retention score) clearly shows me which specific letters a child needs to practice.	[1] [2] [3] [4] [5]
TAM-PU4	Perceived Usefulness	The offline **A4 PDF Progress Report** is useful for sharing a child's phonics progress between home and school without needing internet access.	[1] [2] [3] [4] [5]

Item ID	Construct	Statement	Score (1–5)
TAM-PEOU1	Perceived Ease of Use	The avatar-only profile setup and large 3D buttons allow a 6-to-7-year-old child to start playing without needing to type or read instructions.	[1] [2] [3] [4] [5]
TAM-PEOU2	Perceived Ease of Use	The calm stage backgrounds, clear illustrations, and mouth-shape cues keep the child focused without visual clutter or distraction.	[1] [2] [3] [4] [5]
TAM-PEOU3	Perceived Ease of Use	Running **100% offline** on a standard budget Android phone makes **playIT** practical for real households and classrooms.	[1] [2] [3] [4] [5]
TAM-BI1	Behavioral Intention	I would recommend and use **playIT** as a supplementary phonics learning tool for Grade 1 learners.	[1] [2] [3] [4] [5]

Section 4: Diagnostic Open-Ended Feedback

1. OE-01 **(Observed Strengths):** Which part of **playIT** (Hear It, Say It, Find It, Blend It, Map, or Parent Dashboard/PDF) was most helpful for the learner, and why?
2. OE-02 **(Usability or Pedagogical Friction):** Did you or the learner encounter any confusing screen, audio clip, or recognition issue during the session?
3. OE-03 **(Recommendations for Future Iteration):** What single improvement would make **playIT** more effective for Grade 1 reading remediation?